

KOKKOLA TANKAR, Finland

WMO: 027210

Lat: 63.951N

Lon: 22.844E

Elev: 6

StdP: 101.25

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-17.5	-23.4	0.5	-20.9	-19.5	0.7	-17.4	18.4	0.2	16.8	-1.7	3.6	170	0.817

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.6	22.6	19.2	21.0	18.0	19.6	16.8	19.9	22.2	18.4	20.5	17.2	19.2	4.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.7	21.7	17.5	12.5	19.9	16.3	11.6	18.8	56.8	22.1	52.1	20.8	48.5	19.3	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	14.3	12.8	DB	-21.6	25.5	4.2	2.7	-24.7	27.5	-27.1	29.0	-29.5	30.5	-32.6	32.5	(4)
(5)			WB	-20.9	20.0	5.1	1.9	-24.6	21.3	-27.6	22.5	-30.4	23.5	-34.1	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-5.8	-5.4	-2.8	1.5	7.1	11.7	16.1	15.5	11.8	5.7	1.8	-1.7		
	DBStd	8.44	5.64	6.03	4.33	2.56	3.27	2.94	3.06	2.58	2.54	2.84	3.23	4.76		
	HDD10.0	2445	490	430	398	256	102	14	0	1	13	133	247	362		
	HDD18.3	5006	748	663	657	506	348	200	84	96	196	391	497	620		
	CDD10.0	503	0	0	0	0	12	64	189	169	67	2	0	0		
	CDD18.3	23	0	0	0	0	0	1	15	7	0	0	0	0		
	CDH23.3	36	0	0	0	0	0	2	18	17	0	0	0	0		
	CDH26.7	4	0	0	0	0	0	0	1	3	0	0	0	0		
(14) Wind	WSAvg	6.3	6.8	6.5	6.5	5.9	6.0	5.8	5.2	5.6	6.1	7.0	7.1	7.5		
(15) Precipitation	PrecAvg	453	30	23	23	22	32	40	58	50	50	49	43	36		
	PrecMax	604	61	49	47	62	63	83	117	138	104	87	101	75		
	PrecMin	315	2	0	0	2	1	4	0	15	21	9	0	5		
	PrecStd	80	15	14	12	16	19	22	34	32	27	21	23	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	4.1	6.2	9.2	18.4	21.2	25.2	25.7	19.2	13.6	9.1	5.4		
		MCWB	2.8	2.7	3.0	6.8	13.3	16.0	20.7	20.0	15.9	11.5	8.1	4.6		
	2%	DB	2.5	2.8	4.1	7.5	15.0	18.9	23.3	21.9	17.0	11.4	8.0	4.6		
		MCWB	1.7	1.8	2.3	5.2	11.0	15.2	19.8	18.4	14.7	10.0	7.1	3.8		
	5%	DB	1.6	2.0	3.1	6.0	13.0	17.2	21.7	20.4	15.9	10.3	6.8	3.8		
		MCWB	0.9	1.2	1.8	3.9	10.0	14.5	18.9	17.2	13.9	9.0	6.1	3.0		
	10%	DB	0.8	1.2	2.2	4.8	11.5	15.8	20.4	18.9	15.0	9.5	5.7	3.0		
		MCWB	0.3	0.6	1.0	3.2	8.9	13.4	17.9	16.2	13.4	8.3	5.0	2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.0	3.7	7.6	13.8	17.2	21.5	21.1	16.2	11.7	8.2	4.9		
		MCDB	3.5	3.6	5.1	8.5	17.3	19.8	23.9	24.6	18.2	12.7	8.8	5.1		
	2%	WB	1.9	2.0	2.6	5.5	11.7	15.8	20.5	19.2	15.1	10.2	7.2	4.0		
		MCDB	2.3	2.7	3.7	7.3	14.7	17.9	22.8	21.7	16.6	11.0	7.8	4.6		
	5%	WB	1.1	1.3	1.9	4.2	10.3	14.7	19.3	17.6	14.3	9.4	6.2	3.2		
		MCDB	1.5	1.9	2.9	5.7	12.6	17.0	21.6	19.8	15.6	10.2	6.7	3.7		
	10%	WB	0.4	0.7	1.2	3.4	9.2	13.7	18.1	16.6	13.6	8.5	5.1	2.3		
		MCDB	0.9	1.2	2.0	4.7	11.1	15.6	20.0	18.4	14.8	9.3	5.6	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.3	4.6	3.9	4.5	4.0	3.6	3.8	3.2	2.9	2.8	3.2		
		MCDBR	3.0	3.3	4.3	5.7	7.5	6.2	5.4	6.0	4.5	3.6	3.1	3.4		
	5% WB	MCWBR	2.9	2.9	3.3	3.9	4.7	3.8	3.3	3.2	3.0	3.1	2.9	3.1		
		MCWBR	3.3	3.5	3.9	5.3	7.0	5.5	4.9	5.2	4.2	3.6	3.1	3.4		
(40) Clear-Sky Solar Irradiance	taub		0.225	0.286	0.311	0.350	0.350	0.351	0.371	0.358	0.333	0.314	0.358	0.351		
		taud	2.013	2.142	2.194	2.251	2.442	2.485	2.454	2.483	2.528	2.371	2.592	2.567		
	Ebn at Noon		348	619	769	822	863	868	837	815	764	599	320	167		
		Edh at Noon	28	62	89	106	97	95	96	85	67	51	25	15		
(44) All-Sky Solar Radiation	RadAvg		0.13	0.70	2.22	3.85	5.20	5.77	5.25	3.99	2.26	0.83	0.18	0.05		
	RadStd		0.01	0.14	0.24	0.34	0.51	0.49	0.48	0.37	0.23	0.15	0.03	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page